

W1 – Solving Linear Systems by GRAPHING

Unit 1

MPM2D

1) Solve each system by graphing.

a) $\ell_1: y = 2x + 1$
 $\ell_2: y = x - 2$

Solution: $x = -3, y = -5$

Line 1

$y = 2x + 1$
 slope = $m = \frac{2}{1}$
 y-int = $b = 1$

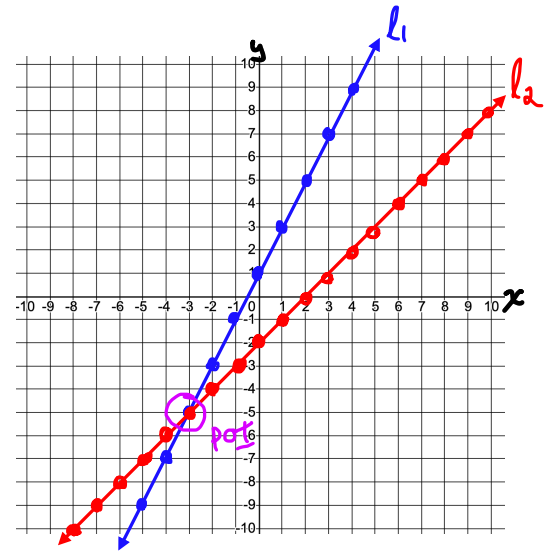
Line 2

$y = 1x - 2$
 slope = $m = \frac{1}{1}$
 y-int = $b = -2$

Check:

ℓ_1
 LS $= y = 2x + 1$
 $= -5 = 2(-3) + 1$
 $= -5 = -5$ ✓

ℓ_2
 RS $= y = x - 2$
 $= -5 = -3 - 2$
 $= -5 = -5$ ✓



b) $\ell_1: x + 2y = 2$
 $\ell_2: x + y = 3$

Solution: $x = 4, y = -1$

Line 1

$x + 2y = 2$
 $2y = -x + 2$
 $y = -\frac{1}{2}x + 1$
 slope = $m = -\frac{1}{2}$
 y-int = $b = 1$

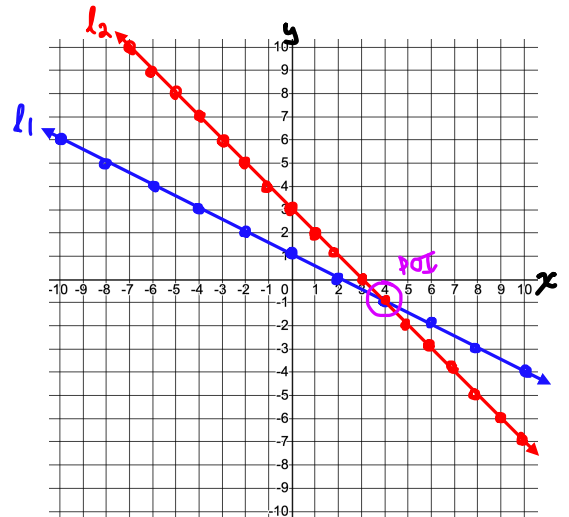
Line 2

$x + y = 3$
 $y = -1x + 3$
 slope = $m = -1$
 y-int = $b = 3$

Check:

ℓ_1
 LS $= x + 2y$
 $= 4 + 2(-1)$
 $= 2$ ✓

ℓ_2
 RS $= x + y$
 $= 4 + (-1)$
 $= 3$ ✓



c) $\ell_1: y = 2x - 3$
 $\ell_2: 2x - y = 5$

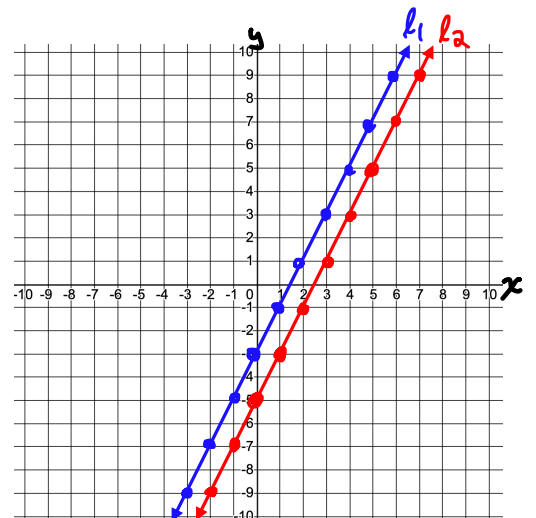
Line 1

$y = 2x - 3$
 slope = $m = \frac{2}{1}$
 y-int = $b = -3$

Line 2

$2x - y = 5$
 $2x - 5 = y$
 $y = 2x - 5$
 slope = $m = \frac{2}{1}$
 y-int = $b = -5$

No solutions. The lines are parallel and distinct



d) $\ell_1: 3x = y + 4$
 $\ell_2: 6x - 2y - 8 = 0$

Line 1

$$3x = y + 4$$

$$3x - 4 = y$$

$$y = 3x - 4$$

$$\text{slope} = m = \frac{3}{1}$$

$$y\text{-int} = b = -4$$

Line 2

$$6x - 2y - 8 = 0$$

$$6x - 8 = 2y$$

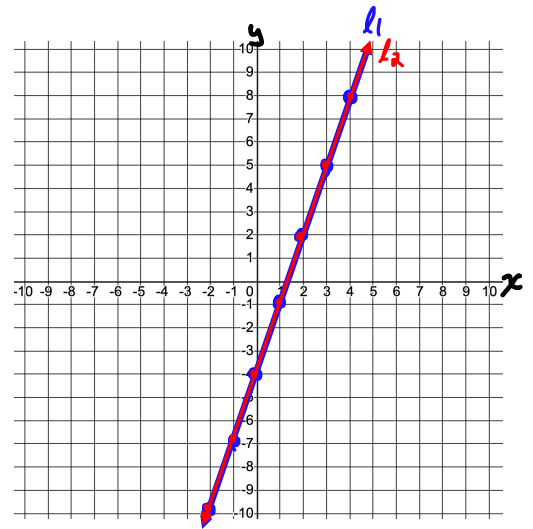
$$3x - 4 = y$$

$$y = 3x - 4$$

$$\text{slope} = m = \frac{3}{1}$$

$$y\text{-int} = b = -4$$

Infinitely many solutions.
 The lines are parallel and coincident.



e) $\ell_1: 3x + 2y = 3$
 $\ell_2: 2x + 10y = -5$

Line 1

$$3x + 2y = 3$$

$$2y = -3x + 3$$

$$y = -\frac{3}{2}x + \frac{3}{2}$$

$$\text{slope} = m = -\frac{3}{2}$$

$$y\text{-int} = b = \frac{3}{2}$$

Line 2

$$2x + 10y = -5$$

$$10y = -2x - 5$$

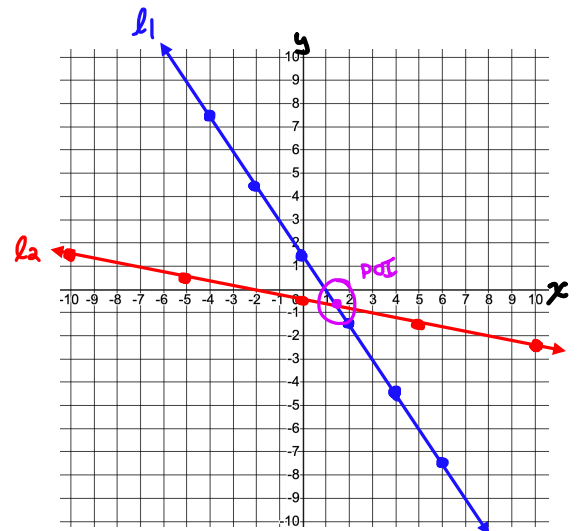
$$y = -\frac{2}{10}x - \frac{5}{10}$$

$$y = -\frac{1}{5}x - \frac{1}{2}$$

$$\text{slope} = m = -\frac{1}{5}$$

$$y\text{-int} = b = -\frac{1}{2}$$

Estimated solution:
 $x = 1.5, y = -0.8$



f) $\ell_1: 2x + 6y - 12 = 0$
 $\ell_2: 6x - 3y - 15 = 0$

solution: $x = 3, y = 1$

Line 1

$$2x + 6y - 12 = 0$$

$$6y = -2x + 12$$

$$y = -\frac{2}{6}x + 2$$

$$y = -\frac{1}{3}x + 2$$

$$\text{slope} = m = -\frac{1}{3}$$

$$y\text{-int} = b = 2$$

Line 2

$$6x - 3y - 15 = 0$$

$$6x - 15 = 3y$$

$$2x - 5 = y$$

$$y = 2x - 5$$

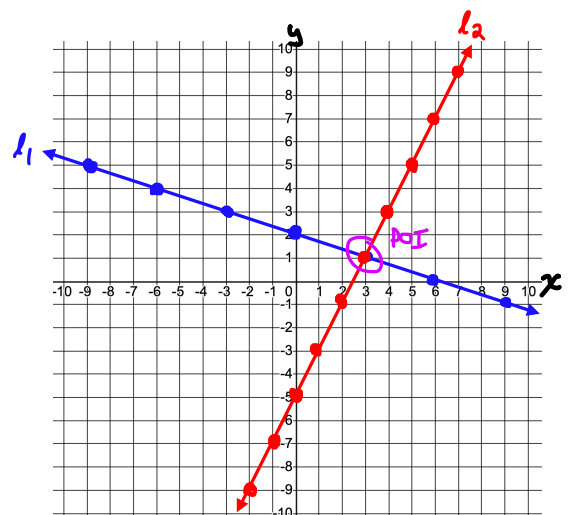
$$\text{slope} = m = \frac{2}{1}$$

$$y\text{-int} = b = -5$$

check:

l1
 $\underline{LS} = 2x + 6y - 12$
 $= 2(3) + 6(1) - 12$
 $= 0$ ✓

l2
 $\underline{LS} = 6x - 3y - 15$
 $= 6(3) - 3(1) - 15$
 $= 0$ ✓



2) Without graphing, determine whether each system has one solution, no solutions, or infinitely many solutions. Explain.

a) $\ell_1: 2x + y = 5$
 $\ell_2: 2x + 10y = -5$

Line 1
 $2x + y = 5$
 $y = -2x + 5$
 slope = $m = -2$
 y-int = $b = 5$

Line 2
 $2x + 10y = -5$
 $10y = -2x - 5$
 $y = -\frac{2}{10}x - \frac{5}{10}$
 $y = -\frac{1}{5}x - \frac{1}{2}$
 slope = $m = -\frac{1}{5}$
 y-int = $b = -\frac{1}{2}$

The lines have different slopes; there is 1 solution to the system.

b) $\ell_1: 3x - y = 0$
 $\ell_2: 6x - 2y = 3$

Line 1
 $3x - y = 0$
 $3x = y$
 $y = 3x + 0$
 slope = $m = 3$
 y-int = $b = 0$

Line 2
 $6x - 2y = 3$
 $6x - 3 = 2y$
 $3x - \frac{3}{2} = y$
 $y = 3x - \frac{3}{2}$
 slope = $m = 3$
 y-int = $b = -\frac{3}{2}$

Same slope but different y-intercepts; the lines are parallel and distinct and have NO solutions.

c) $\ell_1: x + y = 2$
 $\ell_2: 3x = 6 - 3y$

Line 1
 $x + y = 2$
 $y = -x + 2$
 slope = $m = -1$
 y-int = $b = 2$

Line 2
 $3x = 6 - 3y$
 $3y = -3x + 6$
 $y = -1x + 2$
 slope = $m = -1$
 y-int = $b = 2$

Same slope and y-intercept. The lines are parallel and coincident. There are infinitely many solutions.

Answers:

1)a) $(-3, -5)$ b) $(4, -1)$ c) no solutions; parallel and distinct d) infinite solutions; parallel and coincident
 e) $(1.5, -0.8)$ this is an approximate answer f) $(3, 1)$

2)a) one solution b) no solutions c) infinitely many solutions